

single reader is operating in an environment.

2. Multiple-reader environment: The number of simultaneously operating readers is less than the available number of channels.
3. Dense-reader mode: It's the most challenging environment, where the number of readers is greater than the number of channels.

Reader and tag interference can take place within the operating environment—the zone within which the reader's RF signal is attenuated by less than 90 dBc (a radius of approximately 1 km in free space). Consequently, many readers end up operating in a dense-mode environment whether by design or due to neighbouring RFID readers.

A warehouse application with fixed readers and accurate spectrum planning is likely to have minimal interference from neighbours within 1 km. However, a mobile RFID device should expect a dense-reader mode environment due to the lack of control over safe mitigation distances. In this case, it becomes critical to discover what signals may be present in the environment where the RFID system will be, or is, deployed and understand the behaviour of the reader and tags in the presence of interference.

To handle this environment, ISO18000-6C readers that have been certified for dense environments often switch to Miller-modulated sub-carrier (MMS) encoding. This elaborate encoding provides more transitions per bit and is therefore easier to decode in the presence of noise. But it is slower for the same tag's backscatter-link frequency (BLF).

Three different MMS schemes are available: Miller-2, Miller-4 and Miller-8. The number specifies how many BLF periods define a data symbol. For example, using the slowest BLF of 40 kHz, the data rate for Miller-8 is $BLF/8 = 5$ kbps. At such a slow rate, transmitting a 96-bit EPC and 16-bit error check will take 22.4 ms, corresponding to less than 45 tag reads per second (even fewer when all the overhead,

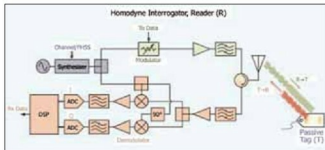


Fig. 1: Block diagram of a typical homodyne interrogator, or tag reader. Using a precision frequency source, the transmitted carrier is modulated and sent to the tag. On the reader's receive side, a single frequency conversion, to base-band, of the backscattered I and Q signals gets processed into the received ID data

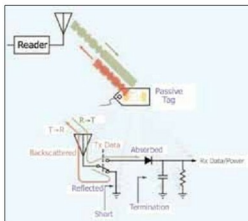


Fig. 2: The passive tag backscatters the interrogator's CW carrier, modulating it by changing the absorption characteristics of the antenna. The passive tag also rectifies the RF energy to create a small amount of power to run the tag

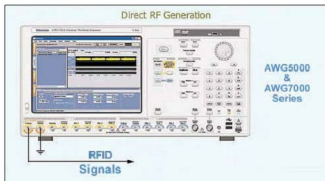


Fig. 3: Arbitrary waveform generators (AWGs) can simulate RFID and other signal interference

such as the forward link commands, is included). Transmitting at this slow rate is not desirable for throughput reasons, but also because some regulations (The United States FCC's Part 15, for instance) allow operation at a single frequency only for an average of 400 ms within a 10- or 20-second period

depending on the 20dB bandwidth of the signal. This regulation may require the tag reader to vacate the channel after 400 ms and hop to a different frequency even if not finished with reading at that frequency.

Readers and tags operating in conformance to ISO18000-7 take a different approach. These use longer RF transmissions with slower transfer rates, which allows the signal to be more immune to interference. For shipping container applications using the commercial equivalent version ISO 18185, this requires that the maximum transmission duration be increased to 60 seconds while maintaining a 10-second minimum silent period between transmissions (FCC part 15.240). At such slow transfer rates, transferring the full 128

kilobytes of data needed to identify all the contents of a shipping container can take up to two minutes. The tags used in accordance to this standard are active; they have an on-board power supply and tend to radiate at higher power levels than passive tags.

Both of these techniques imply that the testing solution needs to collect detailed RF data on pulsed signals over a relatively long time period.

Testing solutions for a dense-mode environment

Simulating a dense-mode environment